

Ceylon Skies:

Building an AI Weather Forecaster for Sri Lanka

The Mission: To create an artificial intelligence system that can "see" current cloud formations over Sri Lanka and accurately predict where they will move and how they will change over the next few hours. This is known as "nowcasting."

1. The Challenge: Why Not Just Use Normal Weather Apps?

Traditional weather forecasting uses massive supercomputers to simulate physics equations over the entire globe. This takes time, meaning the "current" forecast might actually be based on data that is several hours old.

Ceylon Skies takes a different approach. Instead of simulating the entire atmosphere, we use Artificial Intelligence (specifically, Deep Learning) to recognize patterns in live satellite images. It works much like how your brain knows a ball thrown in the air will eventually come down—the AI learns the "physics" of clouds just by observing millions of past examples.

2. Getting the Raw Material: The Data Engineering Phase

Before an AI can learn, it needs a textbook. For Ceylon Skies, that textbook is a massive archive of satellite data.

The Raw Data: We downloaded 8,879 files from the EUMETSAT satellite. These aren't just pictures; they contain precise numerical measurements of "Cloud Top Height"—telling us exactly how tall and thick the clouds are.

However, raw satellite data is messy. It's like trying to read a textbook where the pages are scattered all over the floor. We had to build a complete pipeline to organize it:

- **Targeting Sri Lanka:** Satellites see the whole hemisphere. We carefully extracted only the data points corresponding to Sri Lanka (Latitude: 5.8°N–9.9°N, Longitude: 79.6°E–82.0°E).
- **Fixing the Gaps:** Satellite sensors don't measure perfectly in a grid; they collect scattered points. We used mathematical interpolation to turn these scattered dots into a perfect 256x256 square grid, filling in the blanks.
- **Speaking the AI's Language:** Neural networks prefer numbers between 0 and 1. We normalized cloud heights (ranging from 320 meters to over 15,000 meters) down to this simple scale.

The Result: We successfully processed 8,824 clean files with a 99.4% success rate. More importantly, we checked the timeline and found only 10 missing moments in the entire dataset. This creates a perfect, continuous movie of Sri Lanka's weather.

3. How We Teach the AI: The Training Strategy

Now that we have the data, how does the AI actually learn to predict the future? We use a specialized architecture called a **ConvLSTM** (Convolutional Long Short-Term Memory network).

Think of the ConvLSTM as having two "brains" working together:

- **The "Convolutional" Brain:** This part looks at a single frame and understands spatial relationships. It recognizes the shapes of storm clouds, the coastlines, and where the dense areas are.
- **The "LSTM" (Memory) Brain:** This part remembers what happened in the previous frames. It understands that if a cloud was at point A, and is now at point B, it is moving eastward.

The Training Process (The AI Classroom)

We train the AI by playing a game of "Predict the Future."

1. We show the AI a sequence of past cloud maps (e.g., the last 2 hours).
2. We ask it to draw what the next hour will look like.
3. We compare the AI's drawing to what *actually* happened (since we have the historical data).
4. We score the AI. If it guessed wrong, it adjusts its internal "wiring" slightly to get a better score next time.
5. We repeat this thousands of times until the AI becomes an expert at predicting cloud movement.

4. The Final Product: What the User Sees

The AI does not generate pretty pictures; it outputs pure numbers (density predictions). To make this useful for normal people, Ceylon Skies includes a custom **WebGL Rendering Engine**.

When you visit the website, the backend sends those predicted numbers to your browser. Your computer's graphics card then takes those numbers and smoothly animates them over an interactive 3D map of Sri Lanka, giving you a smooth, real-time forecast that you can scrub through just like a YouTube video.

Current Project Status

Phase	Description	Status
1. Data Engineering	Downloading, extracting, and verifying raw GRIB files from satellites.	100% Complete
2. Dataset Preparation	Cropping to Sri Lanka, regridding into 256x256 arrays, and normalizing data.	100% Complete
3. Quality Assurance	Ensuring temporal consistency (no missing time gaps) for the AI model.	100% Complete
4. AI Model Training	Building the sequence loader and training the ConvLSTM neural network.	Starting Next